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OBTAINED MARK 27

TUTORIAL ANSWER SCRIPT

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Course Code: STAT-251 Course Title: ~~Statistics and~~ Introduction to probability and statistics

Tutorial Examination No: 2 Date: 13/12/2022 Signature of Course Teacher: _____

Ans to the ques no - 3(a)

R = event that she will get job offer

~~P(R) = probability~~

S = event that she will get strong recommendation

M = " " " " good "

W = " " " " weak "

~~P(R) = P~~

$$P(R) = P(R|S).P(S) + P(R|M)P(M) + P(R|W)P(W)$$

$$= (0.8)(0.7) + (0.4)(0.2) + (0.1)(0.1)$$

$$= 0.65$$

Ans to the ques no- 8(b)

For strong recommendation,

$$P(S|R) = \frac{P(R|S)\,P(S)}{P(R)}$$

$$= \frac{0.8 \times 0.7}{0.65}$$

$$= 0.862$$

For moderately good recommendation

$$P(M|R) = \frac{P(R|M)\,P(M)}{P(R)}$$

$$= \frac{0.4 \times 0.2}{0.65}$$

$$= 0.123$$

For weak recommendation

$$P(W|R) = \frac{P(R|W)\,P(W)}{P(R)}$$

$$= \frac{0.1 \times 0.1}{0.65}$$

$$= 0.0154$$

Ans to the ques no- 3(c)

$$P(S|R') = \frac{P(R'|S)\,P(S)}{P(R')}$$

$$= \frac{(1 - P(R|S))\,P(S)}{(1 - P(R))}$$

$$= \frac{(1 - 0.8) \times 0.7}{1 - 0.65}$$

$$= 0.4$$

A

Ans to the ques no - 2(a)

Given,

$$F(x) = \begin{cases} 0 & x < 0 \\ 0.06 & 0 \le x < 1 \\ 0.19 & 1 \le x < 2 \\ 0.39 & 2 \le x < 3 \\ 0.67 & 3 \le x < 4 \\ 0.92 & 4 \le x < 5 \\ 0.97 & 5 \le x < 6 \\ 1 & 6 \le x \end{cases}$$

(i)

the pmf is,

~~for x=0,~~ $p(x=0) = F(0) = 0.06$

$p(x=1) = F(1) - F(0) = 0.19 - 0.06 = 0.$

$p(x=2) = F(2) - F(1) = 0.39 - 0.19 = 0.2$

$p(x=3) = F(3) - F(2) = 0.67 - 0.39 = 0.2$

$p(x=4) = F(4) - F(3) = 0.92 - 0.67 = 0.2$

$p(x=5) = F(5) - F(4) = 0.97 - 0.92 = 0.0$

$p(x=6) = F(6) - F(5) = 1 - 0.97 = 0.0$

$\therefore$ Pmf of x, $p(x) = \begin{cases} 0.06, & x=0 \\ 0.13, & x=1 \\ 0.20, & x=2 \\ 0.28, & x=3 \\ 0.25, & x=4 \\ 0.05, & x=5 \\ 0.03, & x=6 \end{cases}$

(11)

Ans to the ques no:- 2(b)

$$P(2 \leq x < 6) = F(5) - F(1)$$
$$= 0.97 - 0.19$$
$$= 0.78$$

$$P(x > 4) = 1 - P(x \leq 4)$$
$$= 1 - F(4)$$
$$= 1 - 0.92$$
$$= 0.08$$

Ans to the ques no- 2(c)

given, $H(x) = 3 + 2x^2$

x	0	1	2	3	4	5	6
H(x)	3	5	11	21	35	53	75

Now, ~~$E[H(x)] = \sum_{i=0}^{6} x_i p(x_i)$~~

$=$

$$E[H(x)] = \sum_{i=0}^{6} H(x_i)\, p(x_i)$$

$= 0 \times 0.06 + 5 \times 0.13 + 11 \times 0.20 + 21 \times 0.28$

$+ 35 \times 0.25 + 53 \times 0.05 + 75 \times 0.03$

$= 22.38$

$E[(H(x))^2] = 0^2 \times 0.06 + 5^2 \times 0.13 + 11^2 \times 0.20 + 21^2 \times 0.28$

$+ 35^2 \times 0.25 + 53^2 \times 0.05 + 75^2 \times 0.03$

$= 766.38$

$\therefore V[H(x)] = E[(H(x))^2] - (E[H(x)])^2$

$= 766.38 - (22.38)^2$

$= 265.5156$

Ans to the ques no - 1 (a)

We can obtain k from, $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{xy}(x,y)\,dx\,dy = 1$

$\therefore \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} k(2x+y^2)\,dx\,dy = 1$

~~Or, $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}$~~

Or, $k\int_0^3\int_0^2 2x\,dx\,dy + k\int_0^3\int_0^2 y^2\,dx\,dy = 1$

Or, $k\int_0^3 [x^2]_0^2\,dy + k\int_0^3 y^2[x]_0^2\,dy = 1$

Or, $4k\int_0^3 dy + 2k\int_0^3 y^2\,dy = 1$

Or, $4k[y]_0^3 + 2k\left[\frac{y^3}{3}\right]_0^3 = 1$

Or, ~~4k~~ $12k + 18k = 1$

Or, $k = \frac{1}{30}$

Ans to the ques no- 1(b)

$$P(x \le 2, y \le 2) = \int_0^2 \int_0^2 \frac{2x + y^2}{30} \, dx \, dy$$

$$= \frac{1}{30} \int_0^2 \int_0^2 2x \, dx \, dy + \frac{1}{30} \int_0^2 \int_0^2 y^2 \, dx \, dy$$

$$= \frac{1}{30} \int_0^2 [x^2]_0^2 \, dy + \frac{1}{30} \int_0^2 y^2 [x]_0^2 \, dy$$

$$= \frac{4}{30} \int_0^2 dy + \frac{2}{30} \int_0^2 y^2 \, dy$$

$$= \frac{4}{30} [y]_0^2 + \frac{2}{30} \left[\frac{y^3}{3}\right]_0^2$$

$$= \frac{8}{30} + \frac{8}{45}$$

$$= 0.444$$

Ans to the ques no - 1(d)

$$f(x) = \int_0^3 \frac{2x+y^2}{30}\,dy$$

$$= \frac{1}{30}\int_0^3 2x\,dy + \frac{1}{30}\int_0^3 y^2\,dy$$

$$= \frac{6x}{30} + \frac{1}{30}\left[\frac{y^3}{3}\right]_0^3$$

$$= \frac{6x}{30} + \frac{9}{30}$$

$$f(y) = \int_0^2 \frac{2x+y^2}{30}\,dx$$

$$= \frac{1}{30}\left\{\int_0^2 2x\,dx + \int_0^2 y^2\,dx\right\}$$

$$= \frac{1}{30}\left\{4 + 2y^2\right\}$$

$$E[x] = \int_0^2 x f(x)\,dx$$

$$= \frac{1}{30}\int_0^2 6x^2 + 9x\,dx$$

$$= \frac{1}{30}\left[2x^3 + \frac{9x^2}{2}\right]_0^2$$

$$= 1.133$$

$$E[y] = \int_0^3 y f(y)\,dy$$

$$= \frac{1}{30}\int_0^3 4y + 2y^3\,dy$$

$$= \frac{1}{30}\left[2y^2 + \frac{2y^4}{4}\right]_0^3$$

$$= 1.95$$

$$E(xy) = \int_0^3 \int_0^{xy} \frac{2x+y^2}{30}\, dx\, dy$$

$$= \frac{1}{30}\int_0^3 \int_0^2 2x^2 y\, dx\, dy + \frac{1}{30}\int_0^3 \int_0^2 xy^3\, dx\, dy$$

$$= \frac{1}{30}\int_0^3 y\left[\frac{2x^3}{3}\right]_0^2 dy + \frac{1}{30}\int_0^3 y^3\left[\frac{x^2}{2}\right]_0^2 dy$$

$$= \frac{16}{90}\left[\frac{y^2}{2}\right]_0^3 + \frac{2}{30}\left[\frac{y^4}{4}\right]_0^3$$

$$= \frac{4}{5} + \frac{27}{20}$$

$$= 2.15$$

$$Cov(x,y) = E(xy) - E(x)\,E(y)$$

$$= -0.05935$$

$$E(x^2) = \int_0^2 \frac{6x^3 + 9x^2}{30}\, dx$$

$$= \frac{1}{30}\left[\frac{6x^4}{4} + \frac{9x^3}{3}\right]_0^2$$

$$= 1.6$$

$$E(y^2) = \int_0^3 \frac{4y^2 + 2y^4}{30}\, dy$$

$$= \frac{1}{30}\left[\frac{4y^3}{3} + \frac{2y^5}{5}\right]_0^3$$

$$= 4.44$$

$$V(x) = E(x^2) - (E(x))^2$$

$$= 0.3163$$

$$V(y) = E(y^2) - (E(y))^2$$

$$= 0.6375$$

correlation coefficient, $\rho_{xy} = \frac{Cov(x,y)}{\sqrt{V(x)}\cdot\sqrt{V(y)}} = \frac{-0.05935}{0.3163 \times 0.6375}$

$$= -0.132$$

Ans to the ques no- 1(c)

~~$E[Y|X=1.5] = \frac{E(X \cap Y)}{E(X)}$~~

~~$E[Y|X=1.5] = \int_0^3$~~

$$E[X=1.5] = \int_0^{1.5} \frac{6x^2+9x}{30}\,dx$$

$$= \frac{1}{30}\left[\frac{6x^3}{3} + \frac{9x^2}{2}\right]dx$$

$$= 0.5625$$

$$E[x=1.5, y] = \int_0^3 \int_0^{1.5} xy\,\frac{2x+y^2}{30}\,dx\,dy$$

$$= \frac{1}{30}\int_0^3 \int_0^{1.5} 2x^2y\,dx\,dy + \frac{1}{30}\int_0^3 \int_0^{1.5} xy^3\,dx\,dy$$

$$= \frac{1}{30}\int_0^3 y\left[\frac{2x^3}{3}\right]_0^{1.5} dy + \frac{1}{30}\int_0^3 y^3\left[\frac{x^2}{2}\right]_0^{1.5} dy$$

$$= \frac{2}{120}\left[\frac{y^2}{2}\right]_0^3 + \frac{9}{240}\left[\frac{y^4}{4}\right]_0^3$$

$$= 1.0969$$

$$\therefore E[Y|x=1.5) = \frac{1.0969}{0.5625} = 1.95$$